Arman Motiwala

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EDUCATION

Northeastern University, Khoury College of Computer Sciences — Boston, MA
Master of Science in Artificial Intelligence — *Sep 2025 – Dec 2027*

Mukesh Patel School of Technology Management and Engineering — Mumbai, India
Bachelor of Technology in Artificial Intelligence — *Jul 2021 – May 2025*

TECHNICAL SKILLS

Languages & Tools: Python, SQL, JavaScript, Java, Git, Bash, Jupyter Notebook, Docker
Libraries & Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Hugging Face, FastAPI, Flask, pandas, NumPy
ML & Deep Learning: CNNs, GANs, Transformers, U-Net++, EfficientNet, Transfer Learning, RAG, LLM Fine-Tuning
NLP & Data: Text Classification, Sentiment Analysis, NER, NLTK, spaCy, Milvus Vector DB, MongoDB
Cloud & APIs: OpenAI API, Vertex AI, Google Cloud, RESTful APIs

EXPERIENCE

KNOW LOCTOC PTE. LTD. | *Software Developer Intern* | *Singapore (Remote)* — Jan 2025 – Jul 2025

- Designed a custom multi-image analysis module using Vertex AI and GPT-4o to evaluate visual similarity across 1,000+ product images, achieving **95% accuracy** while coordinating with distributed teams across Singapore and India.
- Built a scalable Feedback Analysis Pipeline using FastAPI and Milvus Vector DB, automating sentiment classification across **5 feedback categories**; achieved **F1-score of 0.87** and **ROC-AUC of 0.91**, reducing manual review time by **60%**.
- Architected a RAG-based Quiz Generation System with FastAPI endpoints and OpenAI LLM integrations, supporting **6 question types** aligned with Bloom's Taxonomy for adaptive learning assessment.
- Collaborated across multinational teams to deploy a Slack-based Translation tool using prompt engineering and multi-language LLM evaluation, improving cross-functional communication for **50+ employees**.

Sequel String AI | *AI/ML Intern* | *India (Remote)* — Apr 2024 – Jul 2024

- Engineered an automated image-to-data pipeline for utility meter readings using OpenCV and RESTful APIs, processing **500+ daily images** and improving data extraction throughput by **40%**.
- Integrated automatic language detection and neural machine translation models, enhancing document accessibility for multilingual users across **3 regional languages**.

PROJECTS

Medical Image Analysis: Bone Fracture & Chest X-Ray Diagnosis | *Deep Learning, Computer Vision* — Oct 2025 – Dec 2025

- Developed a bone fracture detection system using U-Net++ with EfficientNet-B4 encoder and SCSE attention, achieving **61.48% Dice score** and **93.14% classification accuracy** on FracAtlas dataset, surpassing YOLO8s-seg baseline by **4.4%**.
- Built a 4-class chest X-ray classifier (COVID-19, Pneumonia, TB, Normal) using EfficientNet-B3 with transfer learning, achieving **84.92% accuracy** and **0.84 macro-F1**, outperforming VGG-16 baselines with **3x fewer parameters**.
- Implemented custom Dice-Focal-BCE loss function to handle extreme class imbalance (<2% fracture pixels), with weighted sampling and label smoothing for respiratory disease classification.

Auto Assess: NL-to-SQL Platform (IEEE SCEECS 2025) | *Python, NLP, LLM Fine-Tuning* — Jul 2024 – Nov 2024

- Developed a platform generating SQL queries from natural language using fine-tuned Llama 3.1 on Spider dataset, achieving **87% execution accuracy** on WikiSQL benchmark.
- Utilized Hugging Face, PyTorch, and LoRA for model fine-tuning and quantization, with MongoDB for backend storage and Flask for API integration.

Sarcasm Detection Model | *NLP, Deep Learning, Neural Networks* — Jan 2024

- Developed and compared CNN and Sequential Neural Network architectures for sarcasm detection, achieving **80.1% accuracy** with comprehensive hyperparameter tuning and model optimization.

PUBLICATIONS & CERTIFICATIONS

Publication: M. A. Motiwala, A. Joshi, and R. Chapaneri, "Auto-Assess: Automated Evaluation Platform for Enhanced SQL Learning," *IEEE SCEECS 2025*, Bhopal, India. DOI: 10.1109/SCEECS64059.2025.10940103

Certifications: Fundamentals of AI Agents Using RAG and LangChain (IBM), Multimodal RAG using Vertex AI (Google), Machine Learning with Python (IBM)

Achievements: Finalist – Datathon 2.0 (2024), Finalist – Cyber Cypher 3.0 Advanced Level (2024)